Project Publication Plan

Optimal Classification of Low-volume Medical Data with Adaptive Kernel Design

20 April 2026

Goal. The goal of this project is to turn the current thesis work on adaptive kernel design into a publishable medical-imaging paper for low-volume mammographic classification.

Current position. The current work includes kernel-bank selection, kernel composition, rotation-aware kernels, and a final signed radial kernel with learnable radius for low-volume mammographic patch classification. The main missing parts for publication are unified evaluation, stronger baselines, better statistics, and more validation.

Target journals and what is still missing

Journal	Deadline	Missing parts before submission
Medical Image Analysis (MedIA)	Rolling	External or patient-level validation, serious deep-learning and radiomics baselines, confidence intervals, stronger clinical framing.
IEEE Transactions on Medical Imaging (TMI)	Rolling	Clearer methodological novelty, stricter ablations, stronger statistical comparisons, and a fully unified leakage-safe protocol.
Artificial Intelligence in Medicine	Rolling	Strong baselines, repeated evaluation, better discussion of clinical relevance and interpretability.
Computerized Medical Imaging and Graphics (CMIG)	Rolling	Unified final experiments, cleaner heatmap/localization evidence, stronger manuscript organization.
Computer Methods and Programs in Biomedicine (CMPB)	Rolling	Complete benchmark table, reproducible scripts, and polished practical presentation of the method.

Plan and goal.

- Publish the project as a medical-imaging journal paper in 2026.
- Use the signed learnable-radius radial kernel as the main method.
- Finish the missing evaluation, baselines, statistics, and final manuscript packaging before submission.

Main tasks that must be completed

- Freeze one final train/validation/test protocol with no patient or image leakage.
- Re-run the strongest bank-based, composition, and radial models on exactly the same split.
- Add at least two strong baselines: one classical radiomics baseline and one compact deep-learning baseline.
- Add confidence intervals and statistical comparison for AUC and accuracy.
- Keep the signed learnable-radius radial model as the main proposed method and treat unconstrained radial results only as ablations.
- Strengthen localization and interpretability evidence with cleaner heatmaps and explanation of why the radial prior matches lesion morphology.
- Convert the final experiments into clean reproducible scripts and final publication-quality tables and figures.

Timeline

Date	Task
1 May 2026	Finish the deep-learning baseline.
8 May 2026	Finalize the evaluation protocol and the exact main claim of the paper.
15 May 2026	Align the baseline and all metrics with the final benchmark setup.
22 May 2026	Re-run the strongest bank-based, composition, and radial models on one common split.
29 May 2026	Freeze the final benchmark setup and main comparison table.
5 June 2026	Compute confidence intervals for the main metrics.
12 June 2026	Finish the key ablation experiments.
19 June 2026	Complete seed-stability checks and error analysis.
26 June 2026	Finalize localization figures and interpretability outputs.
3 July 2026	Finalize reproducible scripts and publication-ready tables.
10 July 2026	Write the methods and results sections in near-final form.
17 July 2026	Complete the full manuscript draft.
24 July 2026	Finish internal revision and figure polishing.
31 July 2026	Final journal formatting, references, and submission package.
7 August 2026	Submit to MedIA/TMI if the evidence is strong enough; otherwise submit to AI in Medicine or CMIG.